

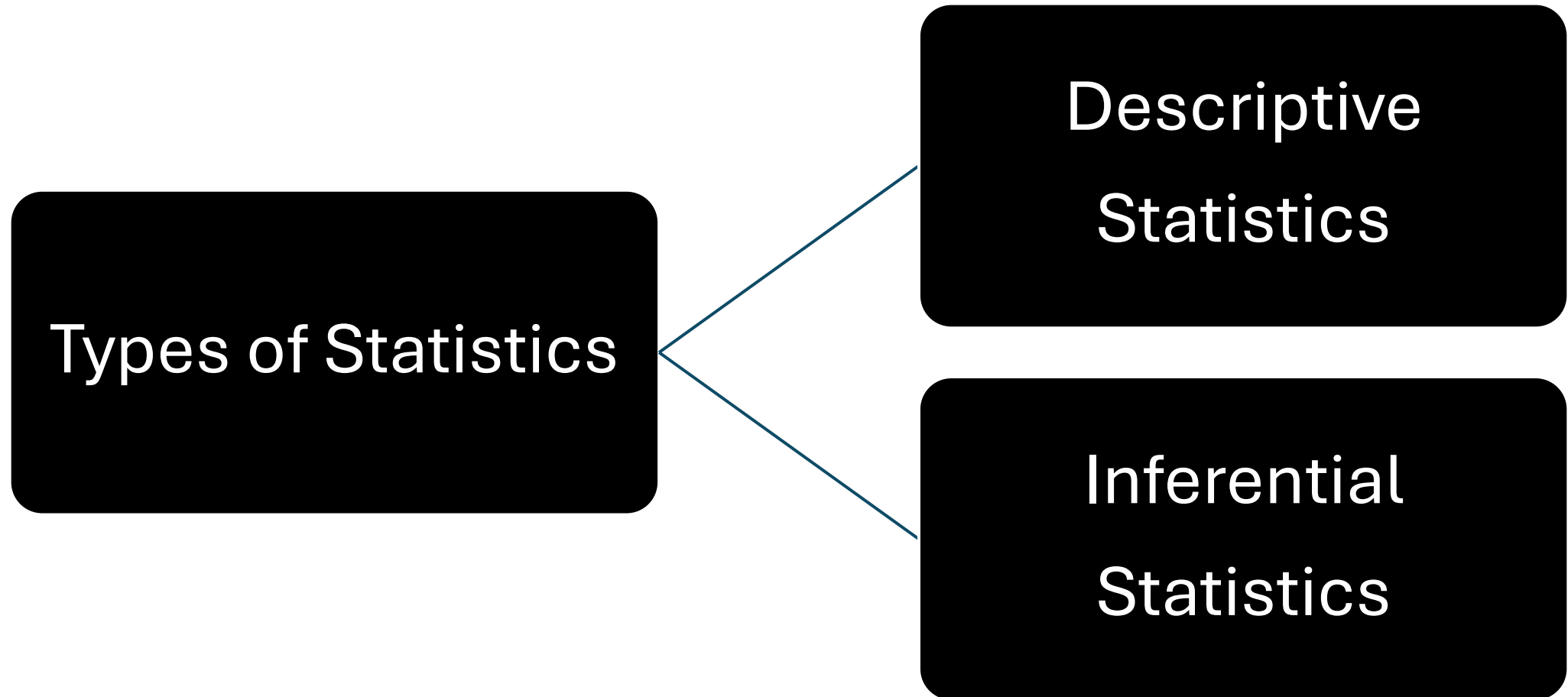
Statistics for Finance

Foundation

Contents

- Descriptive Statistics
- Introduction to hypothesis testing
- Introduction to Stata

Types of Statistics



Types of Statistics

Descriptive Statistics- Involves methods for summarizing, organizing and presenting data so it can be easily understood

Inferential Statistics- Involves making predictions or inferences about population based on sample data.

Descriptive Statistics

- Measures of Central Tendency- *Mean, Median Mode*
- Measures of Dispersion- *Range, Variance, Standard Deviation*
- *Measures of relationship- Correlation & regression*
- Measures of Shape- *Skewness and Kurtosis*
- Data Visualization- *Graphs and Charts*

Measures of Central Tendency

- The **arithmetic mean** (mean) is the most common measure of central tendency.
- The mean can suggest a central value in a set of data
- The mean is the only common measure in which all the values play an equal role.
- Mean is computed by adding together all the values and then dividing that sum by the number of values in the data set.
- The symbol $\bar{X}$, called X-bar, is used to represent the mean of a sample

$$\bar{X} = \frac{\text{sum of the values}}{\text{number of values}}$$

Using the series $X_1, X_2, \dots, X_n$ to represent the set of n values and n to represent the number of values in the sample, the equation becomes

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Measures of Central Tendency

- The **median** is the middle value in an ordered array of data that has been ranked from smallest to largest.
- Extreme values do not affect the median, making the median a good alternative to the mean when such values exist in the data.
- To compute the median for a set of data, you first rank the values from smallest to largest and then use the following formula.

MEDIAN

$$\text{Median} = \frac{n + 1}{2} \text{ ranked value}$$

- In Arithmetic Mean, substituting extreme value changed the mean. Doing the same substitution does not affect the value of median

Measures of Central Tendency

Follow one of the two rules to compute median

- **Rule 1** If the data set contains an *odd* number of values, the median is the measurement associated with the middle-ranked value. •
- **Rule 2** If the data set contains an *even* number of values, the median is the measurement associated with the average of the two middle-ranked values.

Measures of Central Tendency

- The **mode** is the value that appears most frequently.
- Extreme values do not affect mode.
- For a particular variable there can be several or no modes.

Measures of Dispersion/ Variation

- The **range** is the difference between the largest and smallest value
- The range measures the total spread in the set of data.
- Although the range is a simple measure of the total variation of the variable, it does not consider how the values are distributed between the smallest and largest values. Thus, using the range as a measure of variation when at least one value is an extreme value is misleading.

Measures of Dispersion/ Variation

- The range does not consider how the values distribute or cluster between the extremes.
- Two commonly used measures of variation that account for how all the values are distributed are the **variance** and the **standard deviation**.
- It measures the “average” scatter around the mean
- A measure of variation that squares the difference between each value and the mean and then sums these squared differences. The sum of these squared differences, known as the sum of squares (SS), is then used to compute the sample variance (S^2) and the sample standard deviation (S).

Measures of Dispersion/ Variation

SAMPLE VARIANCE

The sample variance is the sum of the squared differences around the mean divided by the sample size minus 1:

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

where

$\bar{X}$ = sample mean

n = sample size

X_i = i th value of the variable X

$\sum_{i=1}^n (X_i - \bar{X})^2$ = summation of all the squared differences between the X_i values and $\bar{X}$

SAMPLE STANDARD DEVIATION

The sample standard deviation is the square root of the sum of the squared differences around the mean divided by the sample size minus 1:

$$S = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

Steps to Calculate Variance and Standard Deviation

1. Calculate difference between each value and the mean
2. Square each difference
3. Sum the squared differences
4. Divide the total by $n-1$ to compute sample variance
5. Take the square root of sample variance to compute sample standard deviation

Measure of relationship

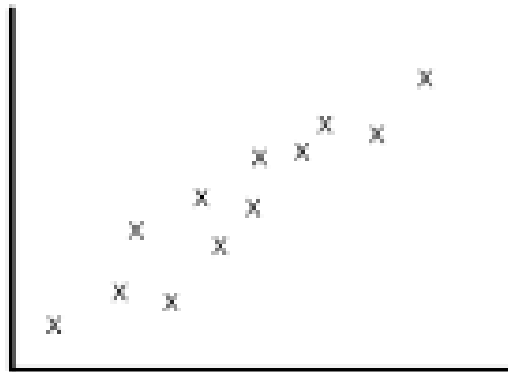
- Correlation describes the relationship between variables in terms of magnitude and direction.
- Correlation coefficient measures the way in which the 'values' in one variable change in relation to 'values' in a second variable.
- The correlation coefficient is measured on a scale of 0 (no correlation) to +1 (perfectly positive correlation) or -1 (perfectly negative correlation).

Coefficient	Weak
$\leq \pm 0.2$	Weak
$\pm 0.3 - 0.6$	Moderate
$\geq \pm 0.7$	Strong

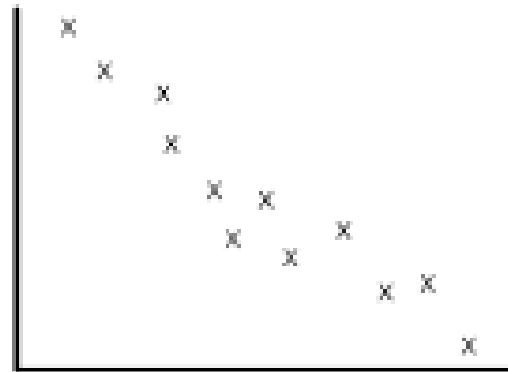
$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Measures of relationship

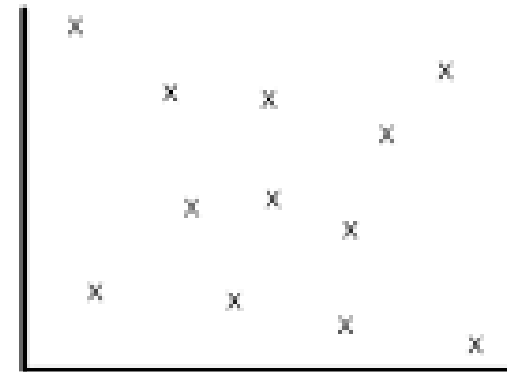
- Correlation between two variables is often presented graphically. Each axis represents a variable; data points are placed along those axes. . Once all of the data have been plotted, the cluster of data points indicates the magnitude and direction of correlation . This type of graph is called a scatterplot.



Positive Correlation



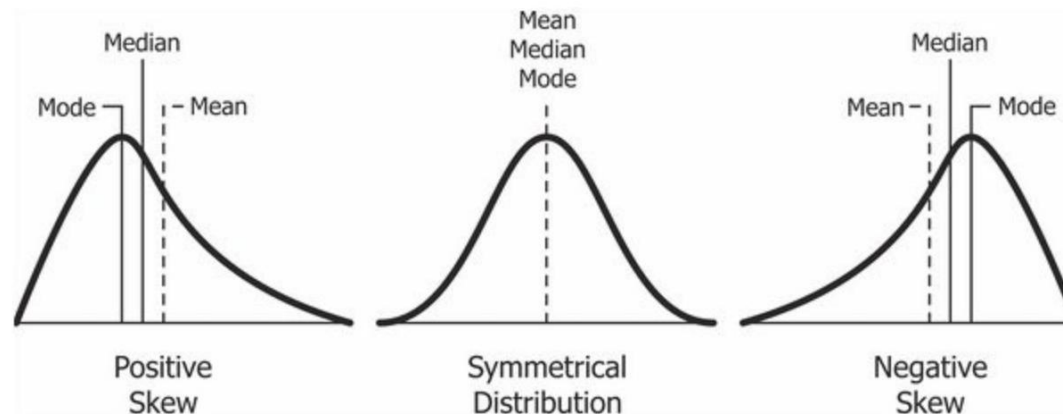
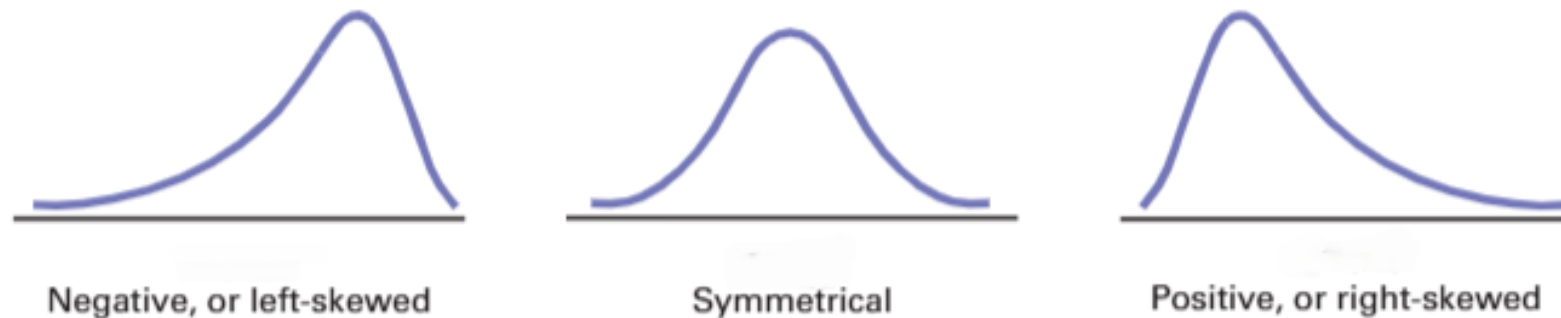
Negative Correlation



No Correlation

Measures of Shape

Skewness measures the extent to which the data values are not symmetrical around the mean.

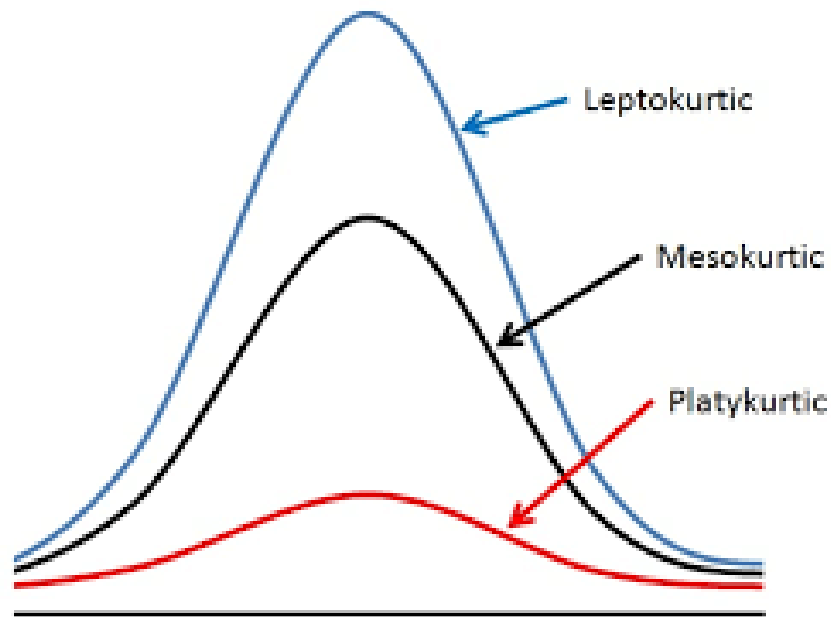


Measures of Shape

- In a **left-skewed** distribution, most of the values are in the upper portion of the distribution. Some extremely small values cause the long tail and distortion to the left and cause the mean to be less than the median
- In a **symmetrical distribution**, values are equally distributed in the upper and lower portions of the distribution. This equality causes the portion of the curve below the mean to be the mirror image of the portion of the curve above the mean and makes the mean equal to the median.
- In a **right-skewed distribution**, most of the values are in the lower portion of the distribution. Some extremely large values cause the long tail and distortion to the right and cause the mean to be greater than the median.

Measures of Shape

- **Kurtosis** measures the peakedness of the curve of the distribution
- Kurtosis compares the shape of the peak to the shape of the peak of a bell-shaped normal distribution, which, by definition, has a kurtosis of zero.



- A distribution that has a sharper-rising center peak than the peak of a normal distribution has positive kurtosis, a kurtosis value that is greater than zero, and is called leptokurtic.
- A distribution that has a slower-rising (flatter) center peak than the peak of a normal distribution has negative kurtosis, a kurtosis value that is less than zero, and is called platykurtic.

Measures of Shape

Leptokurtic

- A distribution that has a sharper-rising center peak than the peak of a normal distribution
- Has positive kurtosis, a kurtosis value that is greater than zero.
- Fatter tail, with more values concentrated on the tail

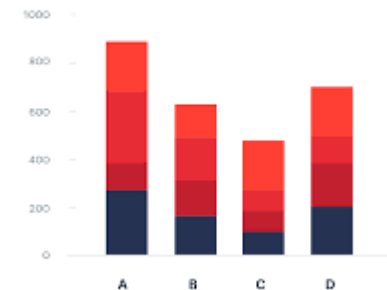
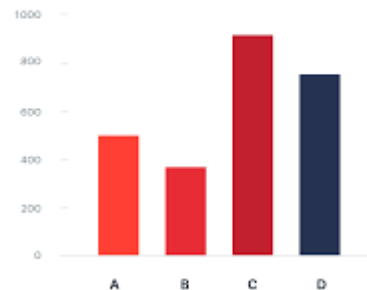
Platykurtic

- A distribution that has a slower-rising (flatter) center peak than the peak of a normal distribution
- Has negative kurtosis, a kurtosis value that is less than zero.
- Flat tail, with fewer extreme values

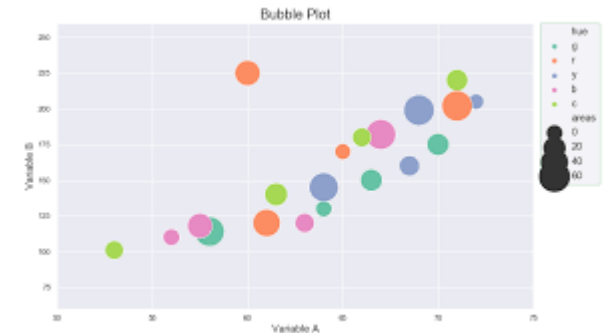
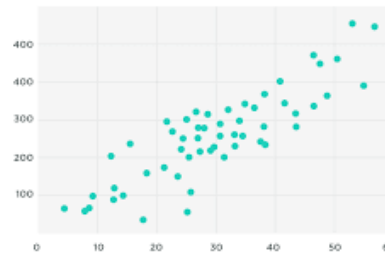
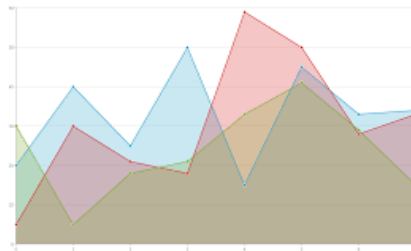
Data Visualisation

- Data visualisation is a general term that describes any attempt to take data and help people to understand it better by presenting it in a visual context.
- Patterns, trends, correlations and other relationships that might otherwise not be noticed in a narrative-type presentation can become clearer and more obvious when presented using some sort of data visualisation software.

- **Pie Chart-** In a pie chart, a single, constant number is represented by the several categories that make up its parts.
- **Donut-** A donut chart is much like a pie chart but with the center area taken out. You can have more sections than a pie chart in a donut chart and it will still be readable.
- **Bar Chart-** Bar charts are used to compare data along two axes. One of the axes is numerical, while the other visualizes the categories
- **Stacked Bar Chart-** A stacked bar chart is a type of bar chart that portrays the compositions and comparisons of several variables.



- **Line Graph-** A line graph is used to show trends, development, or changes through time.
- **Area Chart-** Similar to line chart. However, the space between each line and the X-axis is filled with a specific colour. This indicates a change among different data sets.
- **Scatter Plot-** A scatter plot is a data visualization type used to analyze the correlation between variables. The data is plotted on the chart as dots at the intersection of its two values.
- **Bubble Chart-** Bubble chart is an extension of the scatter plot used to look at relationships between three numeric variables. x-axis and y-axis represents two variables, and the size of the bubbles represents the third variable.



Hypothesis Testing

1. State the hypothesis
2. Select the test
3. Compute test statistic and p-value
4. Compare test statistic or p value to threshold value
5. Conclusion

State the hypothesis

- **Hypothesis**

A hypothesis is an assumption or a statement about the population that may or may not be true.

Hypothesis is tested based on the information obtained from a sample.

- **Null and Alternative Hypothesis**

Null hypothesis is a statement that no difference or relationship exists

Alternative hypothesis is a statement that there exists a difference or relationship

- **Non-directional hypothesis and Directional hypothesis**

A non-directional hypothesis does not specify the direction of difference or relationship.

A directional hypothesis predicts the direction of difference or relationship.

State the hypothesis

- **Type I and Type II error**

The acceptance and rejection of hypothesis is based upon sample result and there is always a possibility that sample not being representative of the population. This could result in error and as a result inferences drawn could be wrong.

	Accept H0	Reject H0
H0 is true	Correct Decision	Type II Error
H0 is false	Type I Error	Correct Decision

Select the test

Parametric and non-parametric test

Parametric	Non-Parametric
Z-test	Wilcoxon signed-rank test
One sample t-Test	
Two or Independent sampled t-Test	Signed-rank test
One-Way Analysis of Variance (ANOVA)	Kruskal-Wallis test

- **Normality test**

Visual inspection- skewness and kurtosis

Statistical tests- Shapiro-Wilk Test and Kolmogorov-Smirnov Test

Select the test

- **Z test**

Compares population mean with sample mean. If sample size is less than 30, one sample t-Test is used

- **Independent sample t-test/ Mann whitney U test**

Compares means of two independent groups

- **Paired sample t-test/ Wilcoxon signed-rank test**

Compares the means of two related groups, before and after group.

- **ANOVA/ Kruskal Wallis Test**

Compares means of three or more groups

- **Chi-square**

Analyses association between two categorical variables

Compute test statistic and p-value

- **test statistic** (calculated from sample data) quantifies how much your sample data deviates from what is expected under the null hypothesis. A large test statistic (either a large positive or large negative value) indicates that the observed result is far from the expected value, suggesting it's less likely to have occurred by chance.
- **p-value** is the probability of observing a test statistic as extreme as, or more extreme than, the one calculated, assuming the null hypothesis is true. A large test statistic leads to a small p value indicating that the observed result are unlikely due to random chance factor and thus, concludes to reject the null hypothesis and vice versa.

Compare test statistic and p-value to critical value and alpha

- **Critical value-** the critical value is compared with the obtained test statistic to determine whether the null hypothesis has to be rejected or not.
 - ✓ *If the test statistic is more extreme than the critical value → Reject the null hypothesis.*
 - ✓ *If the test statistic is less extreme than the critical value → Do not reject the null hypothesis*
- **alpha (significance level)-** the probability of rejecting the null hypothesis when it is true (Type I error). Commonly used alpha are 0.01, 0.05 and 0.10.
 - ✓ If $p\text{-value} \leq \alpha$: The probability of your results occurring by chance is so low (less than your threshold α) that you reject the null hypothesis.
 - ✓ If $p\text{-value} > \alpha$: The probability of your results occurring by chance is higher than your threshold, so you fail to reject the null hypothesis.

Decision

- **Based on comparison of test statistic to critical value**

If the test statistic is more extreme than the critical value, you reject the null hypothesis.

If the test statistic is less extreme than the critical value (e.g., does not fall in the rejection region), you fail to reject the null hypothesis.

- **Based on comparison of p-value to alpha**

If the p-value is less than or equal to alpha ($p \leq \alpha$), you reject the null hypothesis.

If the p-value is greater than alpha ($p > \alpha$), you fail to reject the null hypothesis.

Exercise

You're working as a data analyst for a bank. The marketing department wants to launch a new loan product aimed at customers with an average credit score of 600. To ensure their marketing is accurately targeted, they've asked you to verify this claim.

Your task is to perform a hypothesis test to determine if the average credit score of the bank's customers is indeed 600. Follow these steps:

1. State the hypothesis
2. Interpret the normality test result- (p-value of Shapiro wilks test= 0.25)
3. Determine the appropriate test
4. Analyze the p-value- (test-statistic = 1.8275, p-value=0.076)
5. Formulate a conclusion

Exercise

You're working as a financial analyst. You are assigned with a task to compare the performance of two different investment strategies.

Strategy A: A portfolio that invests in large-cap technology stocks.

Strategy B: A portfolio that invests in a diversified mix of blue-chip stocks.

To test this, you have collected the monthly returns for both portfolios over the same one-year period.

1. State the hypothesis
2. Identify the test (assume normality)
3. Choose a significance level
4. If test statistic result is 2.45 and p-value is 0.011, make a decision and draw conclusion.

Reference

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- Basic Business Statistics- Mark L. Berenson, David M. Levine, Kathryn A Szabat, David F. Stephan
- Research Methods- concepts and cases- Deepak Chawla and Neena Sondhi